

BioSimDR Visualisation Tutorial

BioSimDB (Biomolecular Simulations Database)

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Topology, trajectory, and AiiDA data provenance files from molecular dynamics simulations of biomolecules are stored in a repository specifically designed for their preservation and accessibility.

This resource is part of the **BioSim (Biomolecular Simulations) Data Resources & OPTIMADE Providers' Datasets** resource themes.

How to Access

Quick Links

Visit →

Resource Theme →

1. Go to the BioSimDR trajectory repository and download the trajectory “Human dopamine transporter embedded in a membrane, bound to MDMA”. The trajectories can be found here:
<https://data-collections.psd.ac.uk/communities/biosimdb/records?q=&l=list&p=1&s=10&sort=newest>
2. Load your trajectory file into VMD.
3. Represent the protein in *secondary structure* representation, and colour by *secondary structure*. Smooth the trajectory to make the dynamics clearer.
4. Represent the ligand (drug molecule) in *van der Waals* representation, and colour by *atom name*.
5. Display the *hydrogen bonds* in the protein, and colour them *yellow*.
6. Measure a key inter-atomic distance within your protein and plot a graph of how this changes during the dynamics.
7. Display the lipid membrane as *lines*, and colour by *resname*. What other molecules would have been present in the original simulation, that are not in the slimMD trajectory to reduce its size?
8. Download and explore a different trajectory from BioSimDR. Enjoy!!

Many thanks to Michelle Sahai for donating the human dopamine trajectories, please cite:

<https://doi.org/10.1016/j.pnpbp.2016.11.004>



Combined in vitro and in silico approaches to the assessment of stimulant properties of novel psychoactive substances – The case of the benzofuran 5-MAPB

Michelle A. Sahai^{a,1}, Colin Davidson^{b,c,1}, George Khelashvili^d, Vincenzo Barrese^b, Neelakshi Dutta^b, Harel Weinstein^{d,*}, Jolanta Opacka-Juffry^{a,2}

Faces of HPC: Michelle Sahai

Michelle Sahai is a Senior Lecturer in Biomedical Sciences at the University of Roehampton, London, UK. Michelle's first introduction to HPC was during her undergraduate degree and she is now using HPC resources to simulate membrane proteins.



Institutions:

- University of Roehampton, London, UK
- University of Oxford, Oxford, UK
- University of Toronto, ON, Canada